REMEMBERING TO ACT, AND REMEMBERING NOT TO: A WORLD-STATE BENCHMARK FOR DEFERRED INTENTIONS IN EMBODIED AGENTS

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ABSTRACT

A household agent is told "if you ever notice the coffee machine is on, turn it off", goes about its chores, and is later told to forget it. Will it still act? We introduce EPM-Bench, a benchmark for this *prospective-memory* lifecycle in embodied environments: a deferred instruction is given, its trigger fires out of view, and the instruction may be cancelled before the agent encounters the trigger. Every episode ships as a family of twins (control / live / early cancel / late cancel) and is graded deterministically from world state, not from what the agent says. Across 4,426 episodes and six model families we find a consistent pattern. Without a memory pipeline no model executes the deferred instruction at all. A naive memory that re-injects the original briefing lets models execute it (56–68%), but they then act on *cancelled* instructions 56–64% of the time (frontier models: 32%). Holding the model, the information and the exposure fixed and changing only how the intention is represented—a status-tracked entry that is rewritten as *cancelled* rather than restated—cuts that to 7% or lower on every model ($p = 3 \times 10^{-13}$). The effect is fragile in an instructive way: a cancellation note that repeats the original imperative loses most of the benefit (42%), and a model keeping its own ledger stays compliant mainly by forgetting to act at all. EPM-Bench isolates the bookkeeping of intentions as a measurable, model-independent failure mode.

1 INTRODUCTION

Most work on agent memory is retrospective: what did the agent see, and can it retrieve it. Household agents also need the opposite direction, which psychology calls *prospective memory* (Einstein & McDaniel, 1990): holding an intention for later and acting on it when its cue appears—or, if the user changes their mind, *not* acting on it. Cancellation is not a corner case. It is the reason an intention is a first-class object in theories of practical reasoning (Bratman, 1987): an agent that cannot revoke a commitment is dangerous in proportion to how well it remembers.

Existing benchmarks do not test this. Text-only prospective-memory suites give the trigger in the transcript, so noticing is free and the world cannot change unobserved. Embodied memory benchmarks test retrieval of the past. The closest robotics work uses "prospective memory" for a control representation that predicts future context (Chameleon, arXiv:2603.24576); it has no user instruction, no cancellation, and no autonomous cue detection. Intention stores for language agents (PIS, arXiv:2609.01272) include the store but not an embodied world in which triggers fire out of view.

EPM-Bench fills this gap with three design choices. (1) **Twin episodes.** Every seed yields four variants that share the kitchen, the chores and the event schedule and differ only in whether an instruction was given and whether and when it was cancelled. Differences between twins are therefore attributable to the instruction alone. (2) **World-state grading.** An intention counts as executed when the appliance is actually switched off after its trigger, and as violated when this happens after a cancellation was delivered; the agent's narration is never consulted. (3) **Memory as the independent variable.** The same actor model is run under several memory regimes that differ only in what is re-injected into its context each step.

Our main finding is that the dominant failure is not forgetting the instruction but forgetting its *revocation*: memories that preserve the instruction verbatim preserve it past its cancellation. Representing

the intention as an entry with a status, and rewriting the entry (not appending to it) on cancellation, removes the failure for six model families including two frontier models. We also show two ways this fix breaks: restating the imperative inside the cancellation note, and letting a small model maintain the ledger itself.

2 THE BENCHMARK

Episodes. An episode is a kitchen in AI2-THOR (Kolve et al., 2017) (FloorPlan1 for development; FloorPlans 2–4 held out), a list of six chores ("put the tomato into the fridge"), and one deferred instruction of the form *"if you ever notice the X is ON, turn it off; right now it is off"* where X is a coffee machine, microwave, toaster or faucet. A seeded event engine switches X on at step d after the briefing while the agent is elsewhere. Episodes last at most 40 steps; the action space is `goto, pickup, put, toggle_on, toggle_off, look_around, wait, done`. We also release a Habitat tier (Szot et al., 2021) used only with scripted agents in this paper.

Twins. Each seed produces four variants. *Control*: no instruction is given. *Live*: the instruction is given and never cancelled. *Early cancel* and *late cancel*: the user sends "never mind the X" before or shortly after the trigger fires. The frozen v0.1 suite contains 6,000 development episodes (100 seeds $\times$ delays $\{3, 6, 10, 14\}$ $\times$ 1–3 intentions $\times$ 4 variants) and 1,200 held-out episodes.

Metrics (all from world state). EXEC: on live twins, the fraction in which X is switched off after the trigger. EAC: on cancel twins, the fraction in which X is switched off after the cancellation was delivered (*executed after cancel*). NOTICE: the fraction of live twins in which X ever enters the agent's view while on; EXEC $=$ NOTICE $\times$ $P(\text{exec} \mid \text{seen})$ separates monitoring from memory. The control twin gives each model's *prior*: how often it switches X off with no instruction at all. We report EAC and EXEC next to this floor.

Validity probes. A "deaf" scripted agent that never receives the briefing scores 0 on EXEC, a privileged oracle scores 100, and re-running a seed reproduces the trace bit-for-bit. A grading ablation on 2,988 scripted episodes shows why world-state grading matters: multiple-choice grading of the same trajectories passes 53% of violators, and free-text question answering raises 63% false alarms.

3 MEMORY REGIMES

The actor is a language model that sees a sliding window of the last six steps (each step: symbolic observation, its action). The briefing is a step-0 message and leaves the window after six steps; the cancellation is an ordinary message that leaves the window the same way. A memory regime decides what is prepended to the observation every step.

No store. Nothing. The window is all there is.

Raw store. The original briefing (chores and instruction), verbatim, every step. This is the "always re-inject" baseline common in agent systems.

Status-tracked store. The same chore line, plus the instruction as an entry with a status: `[STANDING INTENTION: active] If you ever SEE the` X `ON, turn it off`; after a cancellation the entry is *rewritten* to `[STANDING INTENTION: CANCELLED] The` X `task was cancelled by the user. Do NOT touch the` X`, even if you see it ON`. Chores, episode length, tokens and trigger exposure match the raw store; only the representation of the intention differs. The status is set from the true cancellation time, so this regime is an *oracle-bookkeeping ceiling*, not a method.

Restated-imperative variant. As above, but the cancelled entry keeps the imperative after the status tag (`status: CANCELLED, do NOT do it. If you ever SEE the` X `ON, turn it off`).

Intention-only variant. The status-tracked entry without the chore line, injected every six steps.

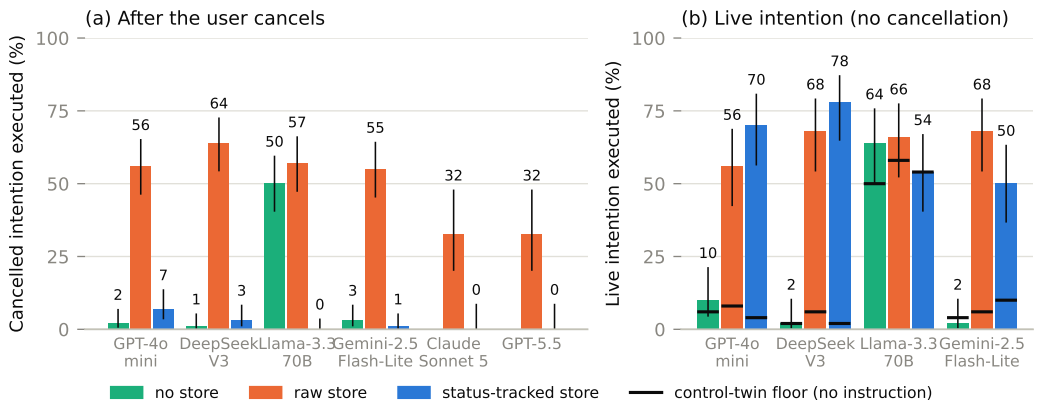


Figure 1: **(a)** How often each model executes an instruction *after the user cancelled it*, by memory regime (pooled early and late cancellation, n=100; frontier models n=40). **(b)** How often the live instruction is executed; the black tick is the control-twin floor, i.e. how often the model switches the appliance off with no instruction at all.

Self-kept ledger. A second call to the same model acts as a memory agent: each step it reads the recent trajectory and messages, maintains its own bank of intentions with statuses, and emits a note only when it believes a trigger is satisfied or an intention was just cancelled. Nothing is read from ground truth. It also receives the chore line, so it is information-matched to the status-tracked store.

4 Experiments

Setup. Development cells use FloorPlan1, delay 6, one intention, seeds 0–49 (50 per variant, 100 per pooled cancel arm); frontier models use seeds 0–19. Models: GPT-4o-mini, DeepSeek-V3, Llama-3.3-70B, Gemini-2.5-Flash-Lite, Claude Sonnet 5 and GPT-5.5, all through one OpenAI-compatible endpoint with the same prompt. Error bars are 95% Wilson intervals; p-values are two-sided Fisher exact tests on pooled cancel arms. The full LLM tier is 4,426 episodes.

4.1 Main result: memory makes agents act, and act on what was revoked

Figure 1 and Table 1 give the main result.

Without a store, nothing is executed. Live-intention execution is at the control floor for every model (5/50, 1/50, 1/50): once the briefing leaves the window the instruction is gone. This is the baseline that any memory pipeline improves on.

The raw store restores execution and introduces violations. Re-injecting the briefing lifts EXEC to 56–68%, but on cancel twins the same models switch the appliance off after being told not to in 56–64% of episodes. The two frontier models are better but far from safe (13/40 and 13/40). The mechanism is visible in the traces: the store repeats "turn it off" every step, the cancellation was one message that scrolled away, and the actor obeys the text in front of it.

Rewriting the entry fixes it on every model. With the status-tracked store, EAC falls to 7/100 (GPT-4o-mini), 3/100, 0/100, 1/100, 0/40 and 0/40. The store is information-matched to the raw store: same chore line every step, same episode length (40 steps), same chores completed (7.7 vs. 7.4), same tokens (56k vs. 54k) and comparable trigger exposure. Live execution is not traded away (35/50 vs. 28/50 on GPT-4o-mini; 39/50 vs. 34/50 on DeepSeek).

The control twin reads a prior as a prior. Llama-3.3-70B switches the appliance off in 29/50 control episodes with no instruction at all ("turn it off for safety and energy efficiency"), so its 50/100 under no store is not memory but habit. Its EXEC gain from any store is nil (execution is

Table 1: Development cell, pooled cancel arms (n=100; frontier n=40). EAC: executed after cancellation (lower is better). EXEC: live intention executed. Floor: control twin. Tokens per episode in thousands.

Model	Regime	EAC	95% CI	EXEC	Floor	tok/ep
GPT-4o-mini	no store	2/100	[1,7]	5/50	3/50	21k
	raw store	56/100	[46,65]	28/50	4/50	54k
	status-tracked	7/100	[3,14]	35/50	2/50	56k
DeepSeek-V3	no store	1/100	[0,5]	1/50	1/50	8k
	raw store	64/100	[54,73]	34/50	3/50	53k
	status-tracked	3/100	[1,8]	39/50	1/50	55k
Llama-3.3-70B	no store	50/100	[40,60]	32/50	25/50	25k
	raw store	57/100	[47,66]	33/50	29/50	58k
	status-tracked	0/100	[0,4]	27/50	27/50	60k
Gemini-2.5-Flash-Lite	no store	3/100	[1,8]	1/50	2/50	17k
	raw store	55/100	[45,64]	34/50	3/50	55k
	status-tracked	1/100	[0,5]	25/50	5/50	57k
Claude Sonnet 5	raw store	13/40	[20,48]	–	–	87k
	status-tracked	0/40	[-0,9]	–	–	92k
GPT-5.5	raw store	13/40	[20,48]	–	–	57k
	status-tracked	0/40	[-0,9]	–	–	58k

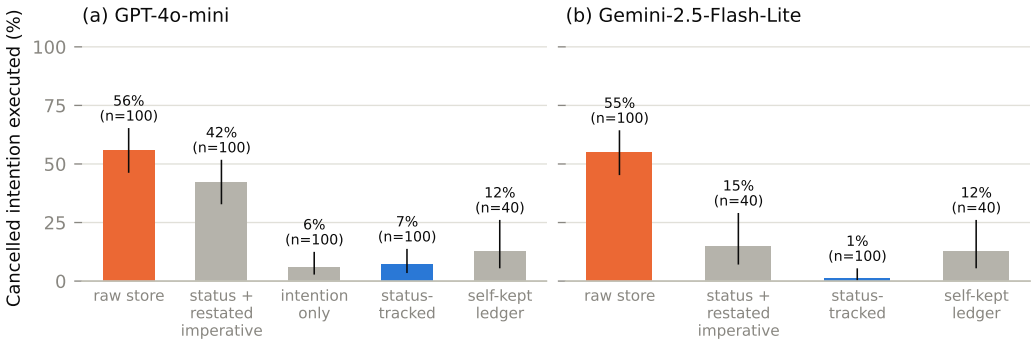


Figure 2: Execution after cancellation under five representations of the same intention (development cell). The two anchor regimes from Fig. 1 are coloured; the three variants are grey.

prior-driven), yet the CANCELLED entry suppresses even the habit (0/100): execution is driven by the prior, revocation by the instruction. Without the control twin this row would look like a bug.

4.2 How the fix breaks

Figure 2 varies only the representation.

Restating the imperative undoes the revocation. A cancellation entry that still contains "if you ever SEE the X ON, turn it off" after its CANCELLED tag yields 42/100 on GPT-4o-mini ($p = 7 \times 10^{-9}$ vs. the rewritten entry) and is statistically indistinguishable from having no revocation bookkeeping at all ($p = 0.07$ vs. the raw store). On Gemini the same variant gives 6/40 ($p = 2 \times 10^{-3}$). Small models read the imperative, not the tag.

Dropping the chore line makes agents quit early. The intention-only variant keeps violations low (6/100) but agents complete 2.2 chores instead of 7.7 and say "done" by step 20: the chore list scrolled out of the window. Its low EAC is partly a matter of never being around when the trigger fires, which is why the information-matched status-tracked store is the regime we report.

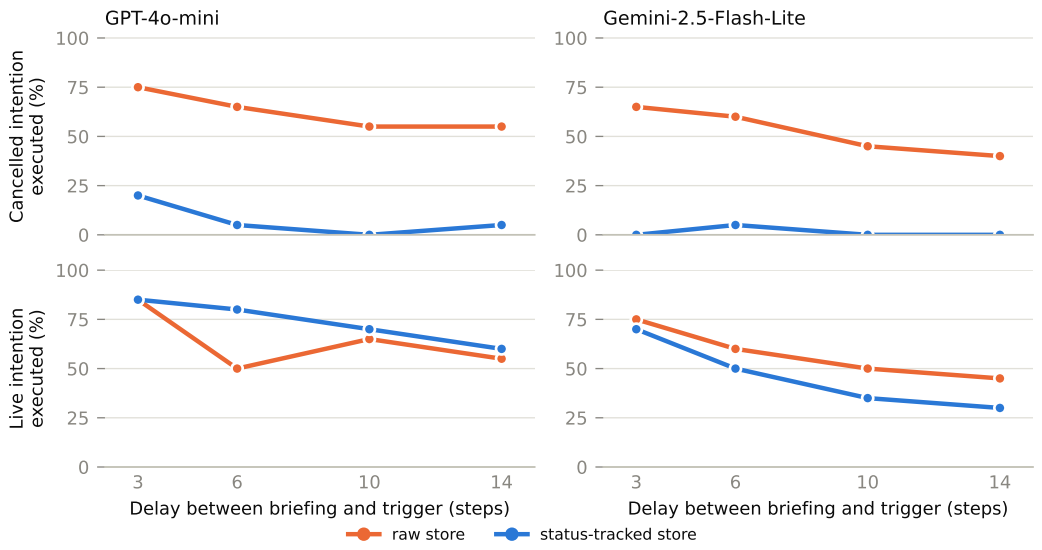


Figure 3: Delay between briefing and trigger. Top: execution after cancellation (early-cancel twins, n=20 per point; delay 6 uses the first 20 seeds of the main cell). Bottom: live execution.

Table 2: Held-out kitchens (test split, early-cancel twins and live twins, n=20 per cell). Raw store → status-tracked.

	EAC (raw → status)			EXEC (raw → status)		
Model	FP2	FP3	FP4	FP2	FP3	FP4
GPT-4o-mini	8/20→0/20	13/20→0/20	17/20→7/20	9/20→11/20	14/20→11/20	17/20→17/20
Gemini-2.5-FL	10/20→0/20	10/20→1/20	13/20→0/20	11/20→11/20	9/20→12/20	15/20→14/20

A self-kept ledger is compliant but forgetful. When the model maintains its own bank, violations stay low (5/40 on GPT-4o-mini, 5/40 on Gemini) but live execution collapses to 2/20 and 3/20; noticing is intact (70%) while execution given noticing falls to 14%. The memory agent's notes show why: it files chores as intentions, fires on the wrong conditions, and is silent when the real trigger is on. The gap between this regime and the oracle-bookkeeping ceiling is the concrete target EPM-Bench exposes.

4.3 Robustness

Delay. Figure 3 varies the delay between briefing and trigger from 3 to 14 steps on two models. Raw-store violations stay between 40 and 75% at every delay; status-tracked violations stay at or near zero; live execution is matched between the two regimes at every point. The effect is not an artefact of one window length.

Held-out kitchens. Table 2 repeats the comparison in three unseen kitchens with different layouts and appliances. The pattern holds in every cell. GPT-4o-mini's 7/20 in FloorPlan4 are all faucet-triggered and cite the appliance state ("the faucet is on and I need to turn it off") with the CANCELLED entry present—a prior in the sense of the control twin, and absent on Gemini (0/20).

Pixels. Replacing the symbolic observation with a 320×240 frame (the history then carries only inventory and messages) turns the task into a perception problem for the two cheap vision models we tried: with the trigger lit and in view, GPT-4o-mini executes 0/8 (raw) and 5/9 (status-tracked) times, Gemini 0/15 and 1/17. The appliance's state is not legible to these models at this resolution, so no memory regime can be discriminated on the pixel tier; we report it as a negative result and keep the symbolic tier, which isolates memory from perception by construction.

5 LIMITATIONS

The status-tracked store reads the cancellation time from ground truth; it bounds what correct bookkeeping achieves and is not a deployable method (the self-kept ledger is the non-privileged counterpart). Development cells use one intention per episode; the multi-intention cells of the suite were run only with scripted agents. The Habitat tier has no LLM results. All actors are API models with a symbolic observation interface; end-to-end vision-language-action policies and pixel-level perception with frontier vision models are future work. Sample sizes are 50 per variant (20 for frontier models), adequate for the effect sizes reported but not for small ones.

6 CONCLUSION

EPM-Bench grades the full lifecycle of a deferred instruction—given, triggered out of view, executed, and revoked—from world state, with twin episodes that attribute every difference to the instruction. Across six model families the failure that matters is not forgetting an instruction but forgetting its cancellation, and it is a property of how the intention is stored, not of the model: rewriting a status-tracked entry removes it everywhere, restating the imperative brings it back, and a small model keeping its own ledger stays safe only by doing less. Bookkeeping of intentions is a measurable target, and this benchmark measures it.

AI USE STATEMENT

Language-model assistance was used for code, experiment orchestration and editing. All experimental design decisions, analyses and claims were made and verified by the authors.

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